STPM Physics · Semester 1 · Chapter 12

# Transfer of Thermal Energy

## 12.1 Conduction, Convection and Radiation

### Definitions & laws

- **Conduction** is the transfer of heat through a material without bulk movement of the material, by lattice vibrations of atoms and, in metals, by the collisions of free electrons; it needs a medium.
- **Convection** is the transfer of heat by the bulk movement of a fluid (liquid or gas) caused by density differences; **radiation** is the transfer of heat by electromagnetic waves (mainly infrared) and does not need a medium.
- **Thermal conductivity** k is the rate of heat flow through a unit area of a material per unit temperature gradient, i.e. the heat transferred per unit time across unit area when the temperature gradient is 1 K $m^{-1}$.
- **Stefan–Boltzmann law**: the total power radiated per unit surface area of a black body is directly proportional to the fourth power of its thermodynamic temperature.
- A **black body** is a perfect absorber and a perfect emitter of radiation at all wavelengths (emissivity e = 1).
- **Steady state**: the temperature at each point does not change with time, so the rate of heat flow is the same through every cross-section in series.

### Formulae

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Rate of heat conduction** | $\frac{Q}{t} = \frac{kA(\theta_1 - \theta_2)}{L} = -kA\frac{d\theta}{dx}$ | $\frac{Q}{t}$ = rate of heat flow (W); k = thermal conductivity (W $m^{-1}$ $K^{-1}$); A = cross-sectional area ($m^2$); L = length or thickness (m); $\theta_1$, $\theta_2$ = temperatures of the two faces; $\frac{d\theta}{dx}$ = temperature gradient (K $m^{-1}$) |
| **Thermal resistance** | $R = \frac{L}{kA}$ | R = thermal resistance (K $W^{-1}$); L = length or thickness (m); k = thermal conductivity (W $m^{-1}$ $K^{-1}$); A = cross-sectional area ($m^2$) |
| **Layers in series** | $R = R_1 + R_2 + \ldots$ | R = total thermal resistance (K $W^{-1}$); $R_1$, $R_2$, … = thermal resistances of the layers |
| **Heat flow through thermal resistance** | $\frac{Q}{t} = \frac{\Delta\theta}{R}$ | Δθ = temperature difference (K); R = thermal resistance (K $W^{-1}$) |
| **Power radiated** | $P = e\sigma AT^4$ | P = power radiated (W); σ = $5.67\times10^{-8}$ W $m^{-2}$ $K^{-4}$; e = emissivity; A = surface area ($m^2$); T = temperature (K, in kelvin) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Net power radiated** | $P_{net} = e\sigma A(T^4 - T_0^4)$ | body at temperature T in surroundings at temperature $T_0$ (K); $P_{net}$ = net power radiated (W); e = emissivity; A = surface area ($m^2$); σ = Stefan–Boltzmann constant |
| **Surface area of a sphere** | $A = 4\pi r^2$ | for a spherical star or planet; r = radius (m); A = surface area ($m^2$) |
| **Intensity at a distance** | $I = \frac{P}{4\pi d^2}$ | I = intensity at distance d (W $m^{-2}$); P = power emitted (W); d = distance from the source (m) |

- Energy balance: at equilibrium, power absorbed = power emitted.

## How to answer (MPM style)

- Conduction: state $\frac{Q}{t} = \frac{kA\Delta\theta}{L}$, then substitute with L in metres and Δθ in K or °C (a difference only), and give the answer in W.
- Composite bar or wall in series: state that the rate of heat flow is the same through each layer in steady state, and write the interface temperature from $\frac{Q}{t}$ equal for the two sections.
- Radiation: convert T to K before using $T^4$, include e (equal to 1 for a black body), and use the $T^4 - T_0^4$ form for net power.
- For a qualitative explanation, link each mode to its mechanism: free electrons (metals, good conductors), density difference (convection currents), and electromagnetic waves (no medium; the Sun heats the Earth through vacuum).

## Section A traps

**Trap:** Using °C in $P = e\sigma AT^4$; the temperature must be in kelvin, but in Δθ for conduction the temperature difference is the same in °C and K.

**Trap:** Stating that the heat flow differs between layers of different materials in series at steady state; the rate of flow is the same, and it is the temperature gradient that is larger in the poorer conductor.

**Trap:** Forgetting the fourth power: doubling the thermodynamic temperature increases the power by a factor of 16, not 2 or 4; and using the radius in place of the area $A = 4\pi r^2$ for a sphere.

## Tricks

**Trick:** Ratios: for the same bar, $\frac{Q}{t} \propto A$ and $\propto \frac{1}{L}$; doubling the radius of a rod gives a 4 times larger heat flow, and doubling the length halves it.

**Trick:** Radiation ratio: $P \propto AT^4$, so compare two bodies with $\frac{A_1}{A_2}(\frac{T_1}{T_2})^4$ and cancel σ and e when they are the same.